

5.1.1 How fast?

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Orders, rate equations and rate constants	
(a) explanation and use of the terms: <i>rate of reaction, order, overall order, rate constant, half-life, rate-determining step</i>	
(b) deduction of: (i) orders from experimental data (ii) a rate equation from orders of the form: $\text{rate} = k[A]^m[B]^n$, where m and n are 0, 1 or 2	M0.2 Learners are expected to interpret initial rates data to determine orders with respect to reactants. Integrated forms of rate equations are not required. PAG10 HSW8 Use of rate equations.
(c) calculation of the rate constant, k , and related quantities, from a rate equation including determination of units	M0.0, M0.1, M0.4, M1.1, M2.2, M2.3, M2.4
Rate graphs and orders	
(d) from a concentration–time graph: (i) deduction of the order (0 or 1) with respect to a reactant from the shape of the graph (ii) calculation of reaction rates from the measurement of gradients (see also 3.2.2 b)	M0.1, M0.4, M1.1, M3.1, M3.2, M3.3, M3.4, M3.5 Concentration–time graphs can be plotted from continuous measurements taken during the course of a reaction (continuous monitoring).
(e) from a concentration–time graph of a first order reaction, measurement of constant half-life, $t_{1/2}$	M3.1, M3.2 Learners should be aware of the constancy of half-life for a first order reaction.
(f) for a first order reaction, determination of the rate constant, k , from the constant half-life, $t_{1/2}$, using the relationship: $k = \ln 2/t_{1/2}$	M0.1, M0.4, M1.1, M2.3, M2.4, M2.5 Learners will not be required to derive this equation from the exponential relationship between concentration and time, $[A] = [A_0]e^{-kt}$.
(g) from a rate–concentration graph: (i) deduction of the order (0, 1 or 2) with respect to a reactant from the shape of the graph (ii) determination of rate constant for a first order reaction from the gradient	M0.1, M0.4, M1.1, M3.1, M3.2, M3.3, M3.4, M3.5 Rate–concentration data can be obtained from initial rates investigations of separate experiments using different concentrations of one of the reactants. Clock reactions are an approximation of this method where the time measured is such that the reaction has not proceeded too far. HSW5 Link between order and rate.

- (h) the techniques and procedures used to investigate reaction rates by the initial rates method and by continuous monitoring, including use of colorimetry (**see also 3.2.2 e**)

PAG9,10

HSW4 Opportunities to carry out experimental and investigative work.

Rate-determining step

- (i) for a multi-step reaction, prediction of,
- (i) a rate equation that is consistent with the rate-determining step
 - (ii) possible steps in a reaction mechanism from the rate equation and the balanced equation for the overall reaction

HSW1 Use of experimental evidence for the proposal of reaction mechanisms.

Effect of temperature on rate constants

- (j) a qualitative explanation of the effect of temperature change on the rate of a reaction and hence the rate constant (**see 3.2.2 f–g**)
- (k) the Arrhenius equation:
- (i) the exponential relationship between the rate constant, k and temperature, T given by the Arrhenius equation, $k = Ae^{-E_a/RT}$
 - (ii) determination of E_a and A graphically using: $\ln k = -E_a/RT + \ln A$ derived from the Arrhenius equation.

M0.3

M0.1, M0.4, M2.2, M2.3, M2.4, M2.5, M3.1, M3.2, M3.3, M3.4

E_a = activation energy,
 A = pre-exponential factor,
 R = gas constant (provided on the *Data Sheet*)
 Explanation of A is **not** required.
 Equations provided on the *Data Sheet*.

HSW5 Link between k and T .